RAG Learning

5.1 Probability

OCR GCSE Mathematics J560 — Unit 5 — Papers 1, 2 and 3 (Foundation and Higher)

5.1.1 Basic Probability Concepts

Probability is a measure of how likely an event is to occur. It is always a value between 0 and 1 (inclusive).

- Probability = 0 → **impossible** (cannot happen)
- Probability = 1 → **certain** (will definitely happen)
- Probability = 0.5 → **equally likely** to happen or not happen

The Probability Scale

Theoretical Probability Formula

When all outcomes are equally likely:

$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}$$

Complementary Events

The **complement** of event A, written A' (or $\bar{A}$), is the event that A does *not* happen.

$$P(A') = 1 - P(A)$$

The probabilities of all mutually exclusive, exhaustive outcomes must sum to 1.

Worked Example: A bag contains 3 red, 5 blue and 2 green counters. A counter is chosen at random.

(a) Find $P(\text{red})$.
Total outcomes = $3 + 5 + 2 = 10$
$P(\text{red}) = \frac{3}{10}$

(b) Find $P(\text{not red})$.
$P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$

(c) Find $P(\text{not blue})$.
$P(\text{not blue}) = 1 - \frac{5}{10} = \frac{5}{10} = \frac{1}{2}$

Exam Tip: Probability can be written as a fraction, decimal or percentage. Fractions are usually preferred in exams unless a decimal or percentage is asked for. Never write probability as a ratio (e.g. 3:10).

5.1.2 Listing Outcomes

For experiments with multiple stages, you must list **all** possible outcomes systematically to avoid missing any.

Sample Space Diagrams

A **sample space diagram** (also called a possibility space) shows all outcomes of two events in a grid.

Worked Example: Two fair dice are rolled. Find the probability that the sum is 7.

Sample space diagram (sum of two dice):

Die 1 \ Die 2	1	2	3	4	5	6
1	2	3	4	5	6	**7**
2	3	4	5	6	**7**	8
3	4	5	6	**7**	8	9
4	5	6	**7**	8	9	10
5	6	**7**	8	9	10	11
6	**7**	8	9	10	11	12

Total outcomes = 36. Outcomes giving sum of 7 (highlighted): 6.

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

Systematic Listing

Worked Example: A coin is flipped and a spinner with sectors 1, 2, 3 is spun. List all outcomes.

Using a tree or ordered list: H1, H2, H3, T1, T2, T3 — **6 outcomes total**.

$P(\text{Head and even number}) = \frac{1}{6}$ (only H2)

Exam Tip: Always count the total number of outcomes carefully before writing any probability. Use a table or systematic list — never guess at the total.

5.1.3 Theoretical vs Experimental Probability

Theoretical probability is calculated from equally likely outcomes using the formula above.

Experimental probability (relative frequency) is calculated from the results of an actual experiment:

$$P(\text{event}) \approx \frac{\text{number of times event occurred}}{\text{total number of trials}}$$

Expected Frequency

$$\text{Expected frequency} = P(\text{event}) \times \text{number of trials}$$

Worked Example: A biased coin is flipped 200 times. It lands on heads 130 times.

(a) Estimate the probability of heads.

$$P(\text{heads}) \approx \frac{130}{200} = 0.65$$

(b) If the coin is flipped 500 more times, how many heads are expected?

Expected heads $= 0.65 \times 500 = 325$

Increasing Reliability

The more trials carried out, the closer the experimental (relative) frequency gets to the true theoretical probability. This is the **law of large numbers**.

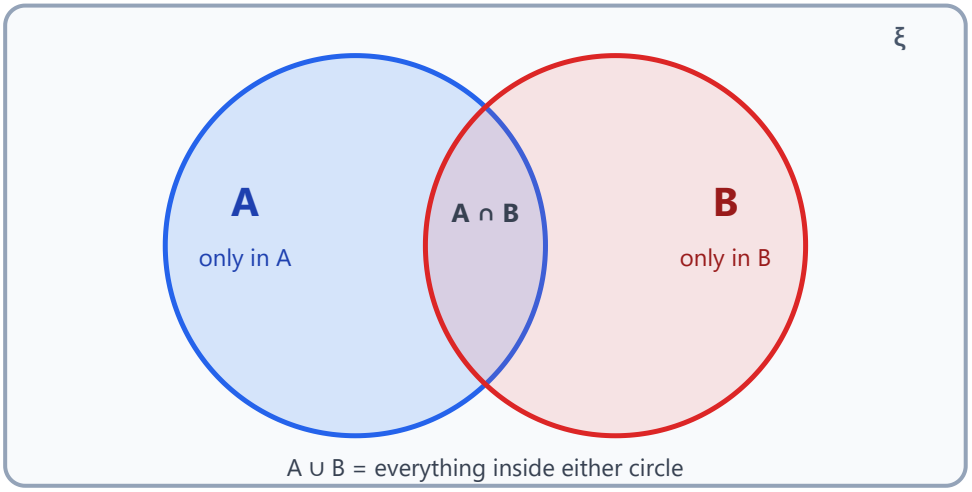
Feature	Theoretical Probability	Experimental Probability
Based on	Equally likely outcomes / logic	Actual results of an experiment
Exact?	Yes (for fair/ideal situations)	An estimate; varies each time
When to use	Fair coins, dice, known situations	Biased items, real-world data

Exam Tip: If you are asked to comment on whether a dice or spinner is fair, compare the experimental probability to the theoretical value $\frac{1}{6}$ (or whatever it should be). The more trials, the more reliable the comparison.

5.1.4 Venn Diagrams

A **Venn diagram** uses overlapping circles within a rectangle (the universal set ξ) to show relationships between sets of outcomes.

- $A \cup B$ — **Union**: outcomes in A *or* B (or both)
- $A \cap B$ — **Intersection**: outcomes in both A *and* B
- A' — **Complement**: outcomes *not* in A



Probability Notation from Venn Diagrams

If the Venn diagram shows frequencies (counts), convert to probabilities by dividing by the total.

Worked Example: In a class of 30 students: 18 study French (F), 12 study Spanish (S), 7 study both.

Step 1 — Fill in the Venn diagram:
Only French: $18 - 7 = 11$ | Both: 7 | Only Spanish: $12 - 7 = 5$ | Neither: $30 - 11 - 7 - 5 = 7$

Step 2 — Find probabilities:

$$P(F \cup S) = \frac{11 + 7 + 5}{30} = \frac{23}{30}$$

$$P(F \cap S) = \frac{7}{30}$$

$$P(\text{neither}) = \frac{7}{30}$$

$$P(F \text{ only}) = \frac{11}{30}$$

Mutually Exclusive Events

Two events are **mutually exclusive** if they cannot both happen at the same time — their circles do not overlap.

If A and B are mutually exclusive: $P(A \cap B) = 0$ and $P(A \cup B) = P(A) + P(B)$

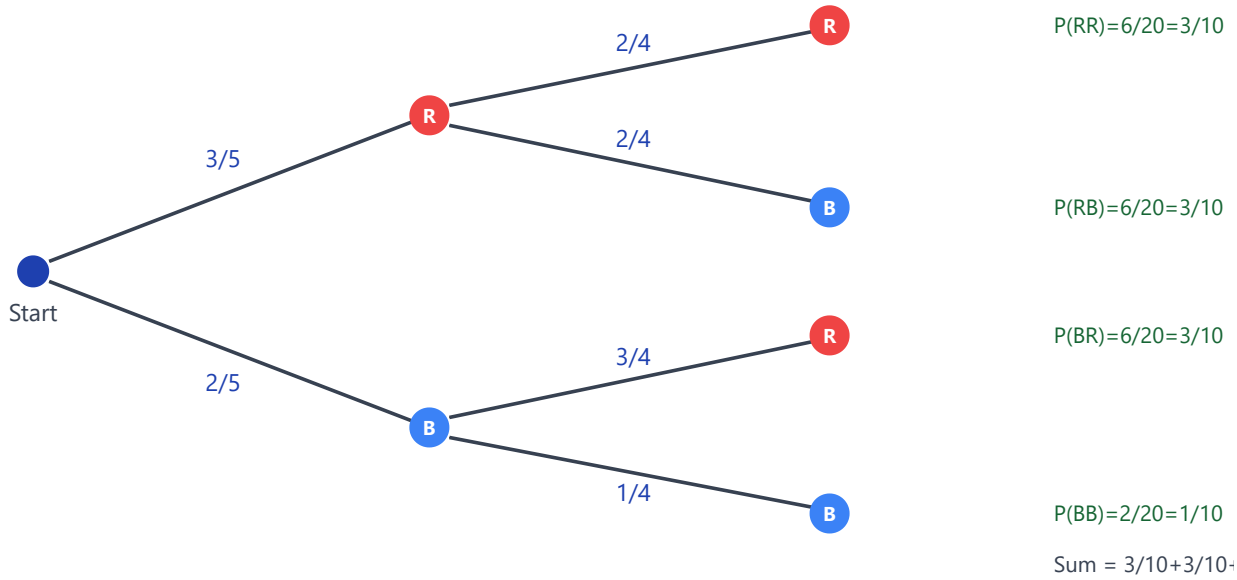
Exam Tip: Always check your Venn diagram frequencies add up to the total before calculating probabilities. It is a very common error to double-count the intersection.

5.1.5 Tree Diagrams

A **tree diagram** shows all possible outcomes of two or more successive events.

- **Multiply** along branches to find the probability of a combined outcome.
- **Add** the probabilities of branches that give the same overall outcome.
- Probabilities on branches from each node must sum to 1.

Tree Diagram: Without Replacement (3 Red, 2 Blue — 5 counters total)



Worked Example: A bag has 3 red and 2 blue counters. Two are drawn **without replacement**.

Find P(one of each colour).

$$P(RB) = \frac{3}{5} \times \frac{2}{4} = \frac{6}{20} \text{ and } P(BR) = \frac{2}{5} \times \frac{3}{4} = \frac{6}{20}$$

$$P(\text{one of each}) = \frac{6}{20} + \frac{6}{20} = \frac{12}{20} = \frac{3}{5}$$

With Replacement vs Without Replacement

With Replacement	Without Replacement
Item is returned before next draw	Item is NOT returned
Probabilities stay the same each stage	Probabilities change at each stage
Events are **independent**	Events are **dependent**
Denominator stays constant	Denominator decreases by 1 each time

Exam Tip: In tree diagrams, always check that the branches from each node sum to 1. Also check that all final outcome probabilities sum to 1 as a verification step.

5.1.6 Conditional Probability [Higher]

Higher Tier Only: **Conditional probability** is the probability of event A given that event B has already occurred.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

This is particularly useful with Venn diagrams and two-way tables.

Worked Example: In a group of 50 people: 30 drink tea (T), 20 drink coffee (C), 10 drink both.

Find P(drinks tea | drinks coffee).

$$P(T \cap C) = \frac{10}{50} = \frac{1}{5}$$

$$P(C) = \frac{20}{50} = \frac{2}{5}$$

$$P(T \mid C) = \frac{P(T \cap C)}{P(C)} = \frac{1/5}{2/5} = \frac{1}{2}$$

Alternative method: Of the 20 coffee drinkers, 10 also drink tea: $\frac{10}{20} = \frac{1}{2}$ ✓

Two-Way Tables

Two-way tables allow you to read off conditional probabilities directly by restricting to the relevant row or column.

Worked Example:

	Passed	Failed	Total
Revised	24	6	30
Did not revise	8	12	20
Total	32	18	50

Find P(passed | revised).

Looking only at the "Revised" row: 24 out of 30 passed.

$$P(\text{passed} \mid \text{revised}) = \frac{24}{30} = \frac{4}{5}$$

Exam Tip: For conditional probability from a two-way table, restrict your attention to only the relevant row or column — that becomes your new "total". Do not use the overall total.

Independent Events

Events A and B are **independent** if $P(A \mid B) = P(A)$ — knowing B occurred gives no information about A.

For independent events: $P(A \cap B) = P(A) \times P(B)$